

Gautham Bellamkonda

+91 8074145535 | gauthambellamkonda@gmail.com |  gautham-bellamkonda | gauthambellamkonda.github.io

RESEARCH INTERESTS

Complexity Theory: classical computational complexity; arithmetic and boolean circuits; boolean function complexity; communication complexity; fine-grained complexity.

Machine Learning Theory: computational learning theory; differential privacy; characterizing expressivity of modern architectures using complexity theory.

Quantum Computing: quantum speedups for classical problems; quantum query complexity; polynomial and adversary methods for finding upper and lower bounds on quantum query complexity.

EDUCATION

Indian Institute of Technology Hyderabad
B.Tech in Computer Science and Engineering, GPA: 8.95/10

Hyderabad, India
2020-2024

PROFESSIONAL EXPERIENCE AND INTERNSHIPS

KLA 
ML Research Intern, Algorithms Engineer

Chennai, India
Jan 2023 – Jul 2023, Jul 2024 – Present

StoryXpress 
ML Research Intern

Hyderabad, India
May 2022 – Jul 2022

RESEARCH PROJECTS

Logic Enhanced Concept Bottleneck Models

Advisor: Vineeth N Balasubramanian

IIT Hyderabad
Apr 2024 – Nov 2024

- Developed a framework integrating symbolic logic reasoning with concept bottleneck models to enhance interpretability and controllability in concept bottleneck classifiers.
- Conducted experiments on standard datasets, demonstrating improved performance and robustness against spurious correlations compared to traditional CBMs.

Lower and Upper Bounds in Quantum Query Complexity

Advisor: KartEEK Sreenivasaiah

IIT Hyderabad
Jan 2024 – May 2024

- Investigated quantum query lower bounds using the adversary method, deriving new insights based on Ambainis's techniques.
- Studied how combinatorial graph structures inform the design and efficiency of quantum algorithms for a few graph problems.

PUBLICATIONS

- Deepika Vemuri, Gautham Bellamkonda, Vineeth N Balasubramanian. **LogicCBMs: Logic-Enhanced Concept-Based Learning**. Accepted at WACV 2026, Arizona, USA. arXiv

TEACHING EXPERIENCE

- **NPTEL Online Course:** Theory of Computation (Spring 2024)
- **IIT Hyderabad:** Discrete Mathematics (Fall 2021), Database Management Systems (Spring 2022), Computer Architecture (Fall 2022), Foundations of Machine Learning (Fall 2023)

SELECTED AWARDS AND HONORS

- Secured a Silver Medal in Math Bowl at the Inter-IIT Tech Meet 2023, representing IIT Hyderabad
- JEE Advanced: Ranked 419 in General Category out of 1,50,000+ candidates in 2020
- JEE Mains: Attained a rank of 243 in General Category out of 10,00,000+ candidates in 2020
- Qualified Indian National Mathematical Olympiad (INMO) in 2019 & 2020. Consequently attended IMOTCs.
- KVPY SA Fellow with AIR 53 out of 1,00,000+ candidates in 2019
- NTSE Scholar: Qualified Stages 1 & 2 of National Talent Search Examination in 2018

REFERENCES

• **Prof. Vineeth N Balasubramanian**
Professor
Department of Computer Science
Indian Institute of Technology, Hyderabad
Email: vineethnb@cse.iith.ac.in
Relationship: B.Tech. Project Supervisor

• **Dr. KartEEK Sreenivasaiah**
Assistant Professor
Department of Computer Science
University of Liverpool
Email: kartEEK.sreenivasaiah@liverpool.ac.uk
Relationship: B.Tech. Project Supervisor